

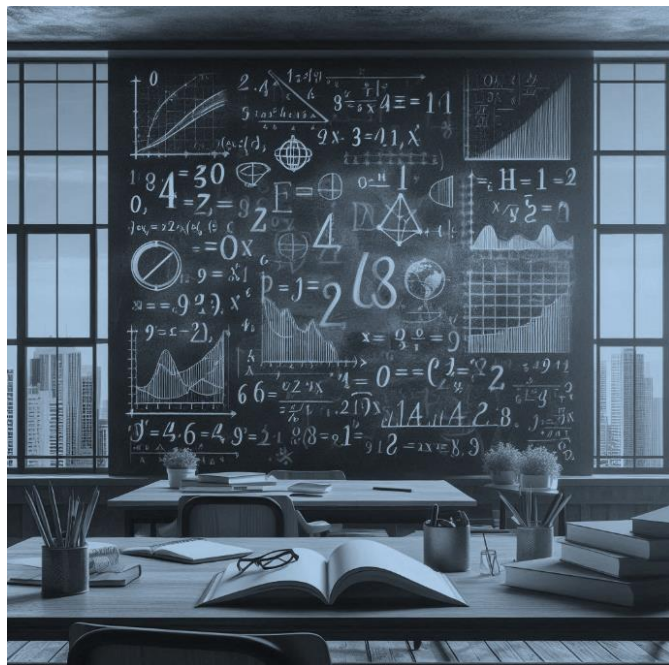


Why are AHP results not binding for appraising real estate?

Miguel Camacaro Pérez, M.Sc

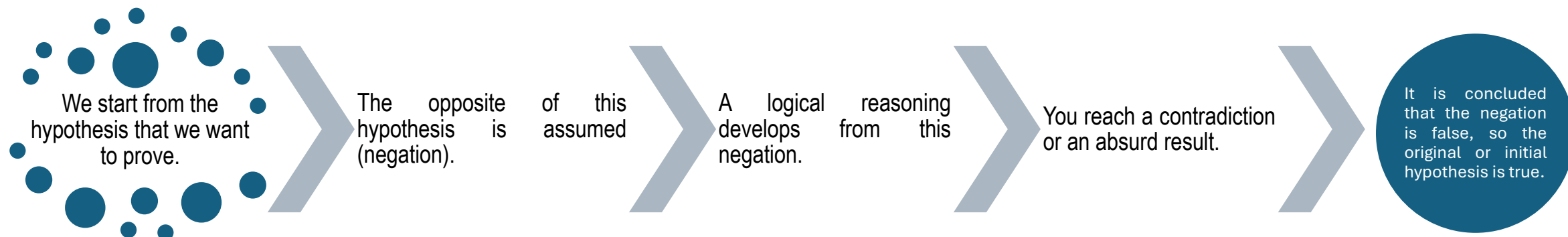


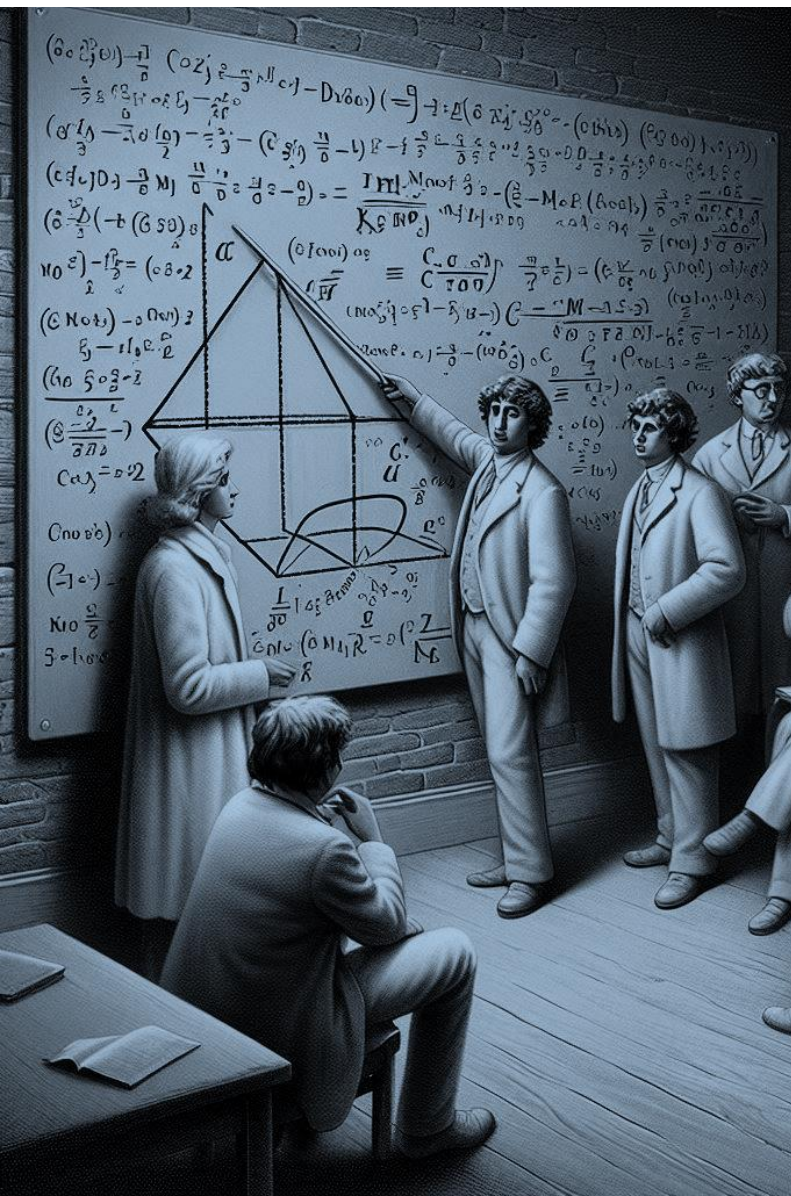
Method of reduction to absurdity



Reduction to absurdity (or *reductio ad absurdum*) is a technique of argumentation and demonstration used to establish the truth of a proposition or situation by showing that assuming the opposite leads to a contradiction or an absurd result. It can be applied in various fields, mainly in logic, mathematics, philosophy and science, but also in everyday situations and legal argumentation.

It consists of temporarily assuming the opposite of what one wants to prove, and then demonstrating that this assumption leads to a contradiction or an absurd result. This implies that the initial assumption is false, therefore, the original statement must be true.





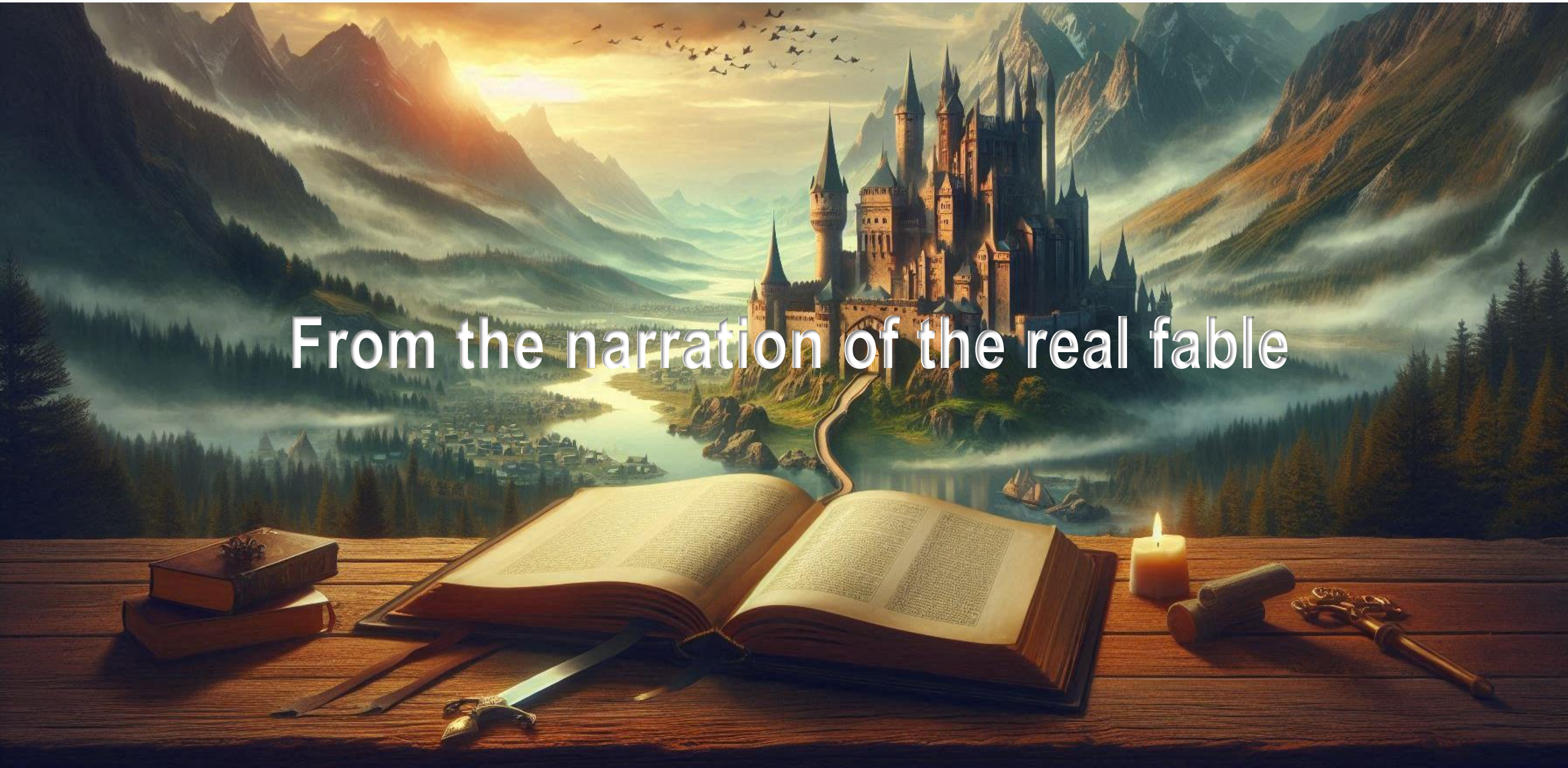
Method of reduction to absurdity

Initial hypothesis (original):

The results of the general weighting vector of the AHP are not linked to real estate market prices.

Alternative hypothesis (contrary):

The results of the general weighting vector of the AHP are linked to real estate market prices.



From the narration of the real fable

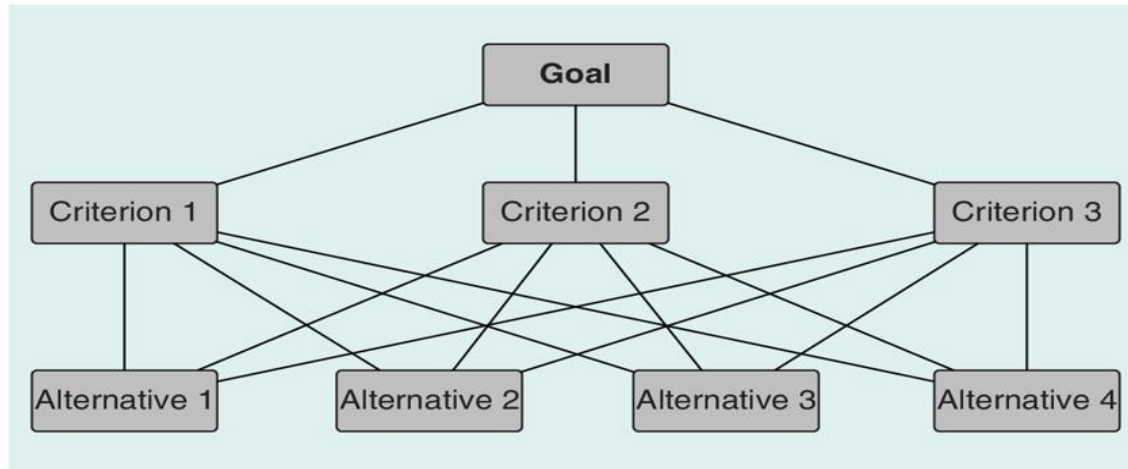


What would be the price of castle N?





ANALYTIC HIERARCHY PROCESS (AHP)



Plus

Valuation Ratio





From the Analytic Hierarchy Process (AHP)

From the origin, the concept and the objective

The Analytic Hierarchy Process (AHP) is a Multi-Criteria Decision Making technique proposed by Thomas Saaty (1977, 1980), classified within the Multiple Attribute Decision Making (MADM) methods, which combines tangible and intangible aspects to obtain, on a ratio scale, the priorities associated with the problem alternatives.

The main characteristics of this approach are the modeling of the problem through a hierarchical structure, the use of pairwise comparisons to incorporate decision-maker preferences, and the obtaining of a ratio scale that is valid for complex decision-making.

From MADM applicable to problems

Selection	Categorization	Sorting	Description
AHP	AHP	AHPSort	
ANP	ANP		
MUAT/UTA	MUAT/UTA	UTADIS	
MACBETH	MACBETH		
PROMETHEE	PROMETHEE	FlowSort	GAIA, FS-GAIA
ELECTRE I	ELECTRE III	ELECTRE-Tri	
TOPSIS	TOPSIS		
Goal Programming			
DEA	DEA		

Source: Ishizaka&Nemery (2013) Multi-Criteria Decision Analysis Methods and Software

The background of the slide is a cinematic image of a king with a long white beard, wearing a golden crown and a blue robe with gold embroidery and a red fur collar. He stands on a stone balcony, looking out over a lush green valley. In the distance, a large, multi-towered castle with blue roofs sits atop a hill. The scene is bathed in the warm, golden light of a low sun, creating a dramatic and majestic atmosphere.

From the solution using Analytic Hierarchy Process (AHP)

Goal

**Select the
Castle**

Criteria

Quality of the environment

Quality of the castle

State of conservation

Alternatives

Castle N° 1

Castle N° 2

Castle N° 3

Castle N° 4

Castle N° 5

Castle N

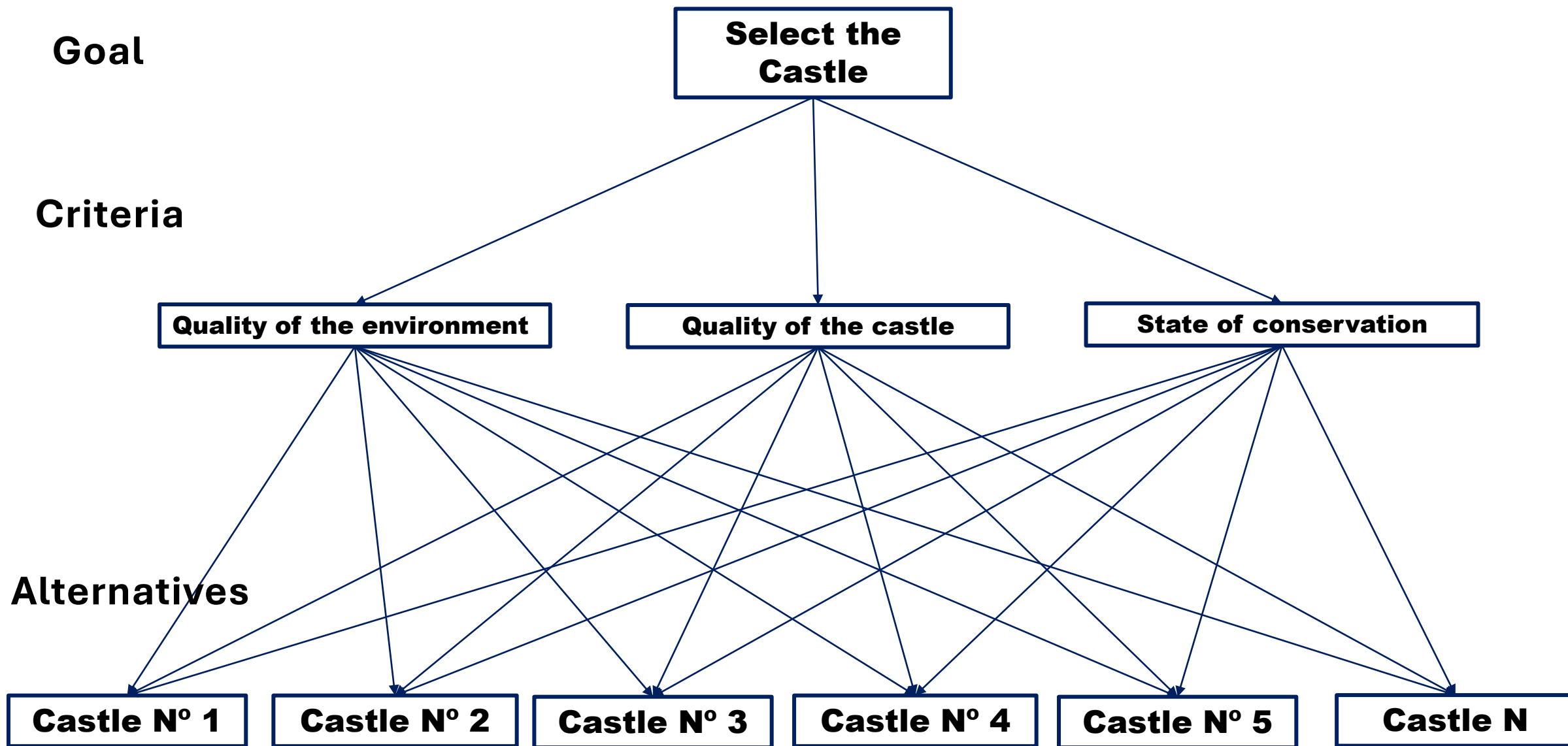


Table 2: Reciprocal matrix of criteria and priority vector¹³

Criteria	Quality of the environment	Quality of the castle	State of conservation	Vector priority
Quality of the environment	1	1/3	1/4	0,12196
Quality of the castle	3	1	1/2	0,31962
State of conservation	4	2	1	0,55842
Consistency ratio =	1,76 %			

Table 3: Reciprocal matrix of the alternatives for the quality of the environment and normalized priority vector

Quality of the environment	Castle 1	Castle 2	Castle 3	Castle 4	Castle 5	Castle N	Normal priority vector
Castle 1	1	3	3	1/3	1/5	1/3	0.0825
Castle 2	1/3	1	1	1/7	1/9	1/7	0.0328
Castle 3	1/3	1	1	1/7	1/9	1/7	0.0328
Castle 4	3	7	7	1	1/3	1	0.2069
Castle 5	5	9	9	3	1	3	0.4381
Castle N	3	7	7	1	1/3	1	0.2069

Consistency ratio = 1.96 %

Table 4: Reciprocal matrix of the alternatives for the quality of the castle and normalized priority vector

Quality of the castle	Castle 1	Castle 2	Castle 3	Castle 4	Castle 5	Castle N	Normal priority vector
Castle 1	1	3	1/3	1	1/5	1	0.0890
Castle 2	1/3	1	1/7	1/3	1/9	1/3	0.0343
Castle 3	3	7	1	3	1/3	3	0.2355
Castle 4	1	3	1/3	1	1/5	1	0.0890
Castle 5	5	9	3	5	1	5	0.4630
Castle N	1	3	1/3	1	1/5	1	0.0890

Consistency ratio = 1.24 %

Table 5: Reciprocal matrix of the alternatives by state of conservation and normalized priority vector

State of conservation	Castle 1	Castle 2	Castle 3	Castle 4	Castle 5	Castle N	Normal priority vector
Castle 1	1	5	3	1	1	3	0.2559
Castle 2	1/5	1	1/3	1/5	1/5	1/3	0.0428
Castle 3	1/3	3	1	1/3	1/3	1	0.0947
Castle 4	1	5	3	1	1	3	0.2559
Castle 5	1	5	3	1	1	3	0.2559
Castle N	1/3	3	1	1/3	1/3	1	0.0947

Consistency ratio = 0.93 %

Table 6: Global vector results and ranking of alternatives

Castle	Quality of the environment	Quality of the castle	State of conservation	Vector priority criteria	Normalized global vector	<i>Ranking</i>
Castle 1	0.08247	0.0890	0.2559	0.12196	0.1814	3
Castle 2	0.03282	0.0343	0.0428	0.31962	0.0389	6
Castle 3	0.03282	0.2355	0.0947	0.55842	0.1322	4
Castle 4	0.20688	0.0890	0.2559		0.1966	2
Castle 5	0.43813	0.4630	0.2559		0.3443	1
Castle N	0.20688	0.0890	0.0947		0.1066	5

As can be seen, Castle 5 was the one that obtained the highest value among the elements of the normalized global vector (0.3443), being number 1 in the ranking. As for Castle N, with a value of 0.1066, it occupied the fifth position in the selection process.

If a decision had to be made, the Castle 5 option should be considered, but this would not necessarily be definitive because the Benefits, Costs, Opportunities and Risks would have to be analyzed for the final decision.

Up to this point, the AHP has been correctly applied in its original form, and for which it was idealized in the 1970s by its author, and today, almost 50 years later, it continues to be studied by the scientific community as an alternative to empirically support complex decision-making.



What is the price of castle N?

Valuation Ratio as a mathematical artifice

$$\text{Valuation Ratio} = \frac{\sum \text{Comparable property prices}}{\sum \text{Weighting of comparable properties}}$$

Unit Value [UM/m ²]
1,650.00
1,200.00
1,575.00
2,040.00
3,100.00

Normalized global vector
0.1814
0.0389
0.1322
0.1966
0.3443
0.1066

Value of property = Valuation Ratio x the weighting of the property being appraised

Table 7: Castle N Unit Value Results based on Valuation Ratio

Castle	Unit Value [UM/m ²]	Vvg [adim]	Valuation Ratio [UM/m ²]	Vvg N [adim]	Unit Value N [UM/m ²]
Castle 1	1,650.00	0.1814	10,706.17	0.1066	1,141.17
Castle 2	1,200.00	0.0389			
Castle 3	1,575.00	0.1322			
Castle 4	2,040.00	0.1966			
Castle 5	3,100.00	0.3443			
Total	9,565.00	0.8934		1.0000	

Table 8: Total value of the Castle N¹⁵

Castle	Construction area [m ²]	Unit Value [UM/m ²]	Total Value [UM]
Castle N	1,500.00	1,150.00	1,725,000.00



Castle 2 is no longer for sale...

Table 9: Normalized priority vector of the alternatives by criteria, priority vector of the criteria, global vector and ranking.

Castle	Quality of the environment	Quality of the castle	State of conservation	Vector priority criteria	Normalized global vector	Ranking
Castle 1	0.0812	0.0876	0.2727	0.12196	0.190210	3
Castle 3	0.0338	0.2364	0.0909	0.31962	0.130446	4
Castle 4	0.2056	0.0876	0.2727	0.55842	0.205383	2
Castle 5	0.4738	0.5007	0.2727		0.370109	1
Castle N	0.2056	0.0876	0.0909		0.103852	5
Consistency ratio =	2.28 %	0.94 %	0.00 %			

Table 11: Total value of Castle N (excluding Castle 2)

Castle	Construction area [m ²]	Unit Value* [UM/m ²]	Total Value [UM]
Castle N	1,500.00	970.00	1,455,000.00

* Rounded

Table 10: Castle N Unit Value Results based on Valuation Ratio (without Castle 2)

Castle	Unit Value [UM/m ²]	Vvg [adim]	Valuation Ratio [UM/m ²]	Vvg N [adim]	Unit Value N [UM/m ²]
Castle 1	1,650.00	0.1902	9,334.40	0.1039	969.40
Castle 3	1,575.00	0.1304			
Castle 4	2,040.00	0.2054			
Castle 5	3,100.00	0.3701			
Total	8,365.00	0.8961		1.0000	

Leave castle 2 and take out castle 4...



Table 12: Normalized priority vector of the alternatives by criteria, priority vector of the criteria, global vector and ranking.

Castle	Quality of the environment	Quality of the castle	State of conservation	Vector priority criteria	Normalized global vector	Ranking
Castle 1	0.1117	0.0983	0.3434	0.12196	0.236831	2
Castle 2	0.0435	0.0388	0.0551	0.31962	0.048475	5
Castle 3	0.0435	0.2527	0.1290	0.55842	0.158130	3
Castle 5	0.5277	0.5119	0.3435		0.419759	1
Castillo N	0.2735	0.0983	0.1290		0.136805	4
Consistency ratio =	2.20 %	0.94 %	2.20 %			

Table 14: Castle N Unit Value Results based on Valuation Ratio (without Castle 4)

Castle	Construction area [m ²]	Unit Value* [UM/m ²]	Total Value [UM]
Castillo N	1,500.00	1,200.00	1,800,000.00

*Rounded

Table 13: Results of the Unit Value of Castle N based on the Valuation Ratio (without Castle 4)

Castle	Unit Value [UM/m ²]	Vvg [adim]	Valuation Ratio [UM/m ²]	Vvg N [adim]	Unit Value N [UM/m ²]
Castle 1	1,650.00	0.2368	8,717.61	0.1368	1,192.61
Castle 2	1,200.00	0.0485			
Castle 3	1,575.00	0.1581			
Castle 5	3,100.00	0.4198			
Total	7,525.00	0.8632		1.0000	



Include a new offer: the castle 6...

Table 15: Normalized priority vector of the alternatives by criteria, priority vector of the criteria, global vector and ranking.

Castle	Quality of the environment	Quality of the castle	State of conservation	Vector priority criteria	Normalized global vector	Ranking
Castle 1	0.1190	0.1002	0.2168	0.12196	0.1676	2
Castle 2	0.0266	0.0311	0.0342	0.31962	0.0323	7
Castle 3	0.0266	0.1953	0.0853	0.55842	0.1133	5
Castle 4	0.1548	0.0727	0.1817		0.1436	4
Castle 5	0.3435	0.3955	0.2378		0.3011	1
Castle 6	0.1612	0.1215	0.1588		0.1472	3
Castle N	0.1684	0.0837	0.0853		0.0949	6
Consistency ratio =	10.96 %	4.40 %	6.79 %			

Table 17: Results of the Unit Value of Castle N based on the Valuation Ratio (with Castle 6)

Castle	Unit Value [UM/m ²]	Vvg [adim]	Valuation Ratio [UM/m ²]	Vvg N [adim]	Unit Value N [UM/m ²]
Castle 1	1,650.00	0.1676	13,882.92	0.0949	1,317.92
Castle 2	1,200.00	0.0323			
Castle 3	1,575.00	0.1133			
Castle 4	2,040.00	0.1436			
Castle 5	3,100.00	0.3011			
Castle 6	3,000.00	0.1472			
Total	12,565.00	0.9051		1.0000	

Table 18: Total value of Castle N (considering Castle 6)

Castle	Construction area [m ²]	Unit Value* [UM/m ²]	Total Value [UM]
Castle N	1,500.00	1,320.00	1,980,000.00

*rounded

From the comparison of results...



Table 16: Comparison of ranking of alternatives by case

Castle	Original case 6A	Case 5A (-2)	Case 5A (-4)	Case 7A (+6)
Castle 1	3	3	2	2
Castle 2	6	NA	5	7
Castle 3	4	4	3	5
Castle 4	2	2	NA	4
Castle 5	1	1	1	1
Castle 6	NA	NA	NA	3
Castle N	5	5	4	6

Table 19: Comparative results of the unit and total values for each case

Case	Unit Value [UM/m ²]	Total Value [UM]
Original Case 6A	1,150.00	1,725,000.00
Caso 5A (-2)	970.00	1,455,000.00
Caso 5A (-4)	1,200.00	1,800,000.00
Caso7A (+6)	1,320.00	1,980,000.00
Mean	1,160.00	1,740,000.00
Standard Deviation	145.37	218,059.62
Coefficient of Variation	12.53%	12.53%

36.08 %

Non-Ratio Reciprocity Test (NRRT)

Table 20: Value of Castle 1 using the Valuation Ratio (Original Case)

Castle	Unit value [UM]	Vvg [adim]	Valuation ratio [UM/m ²]	Vvg 1 [adim]	Unit Value 1 [UM/m ²]	Unit price [UM/m ²]	NRRT
Castle 2	1,200.00	0.0389	11,063.26	0.1814	2,007.09	1,650.00	22%
Castle 3	1,575.00	0.1322					
Castle 4	2,040.00	0.1966					
Castle 5	3,100.00	0.3443					
Castle N	1,141.17	0.1066					
Totals	9,056.17	0.8186		1.0000			

$$NRRT = \frac{\text{Estimated value} - \text{offered price}}{\text{Offered price}}$$

Table 21: Comparison of the results of the unit, total and NRRT values for each case

Castle	Offers	Original Case		Original Case (-A2)		Original Case (-A4)		Original Case (+1A)	
		Unit Value [UM/m ²]	NRRT [adim]	Unit Value [UM/m ²]	NRRT [adim]	Unit Value [UM/m ²]	NRRT [adim]	Unit Value [UM/m ²]	NRRT [adim]
1	1,650.00	2,007.09	22%	1,804.97	9%	2,193.26	33%	2,326.96	41.03%
2	1,200.00	384.63	-68%	NA	NA	382.98	-68%	422.99	-64.75%
3	1,575.00	1,390.80	-12%	1,164.02	-26%	1,341.61	-15%	1,572.53	-0.16%
4	2,040.00	2,120.60	4%	1,885.37	-8%	NA	NA	1,764.75	-13.49%
5	3,100.00	3,994.44	29%	3,663.18	18%	4,063.90	31%	5,102.38	64.59%
6	3,000.00	NA	NA	NA	NA	NA	NA	1,878.27	-37.39%
N	NA	1,141.17	0%	969.40	0,00%	1,192.61	0%	1,192.05	0,00%



Wise
Analytical

Magnanimous
King

From the opinion of the Wise Analytical

Table 22: Reciprocal matrix of criteria and priority vector

Criteria	Strength	Security	Access	Vector normalized priority
Strength	1	1/4	1/4	0.10852
Security	4	1	2	0.54693
Access	4	1/2	1	0.34454

Consistency ratio= 5,16%

Table 23: Reciprocal matrix of alternatives by strength criterion and normalized priority vector ²⁰

Strength	Castle 1	Castle 2	Castle 3	Castle 4	Castle 5	Castle N	Normal priority vector
Castle 1	1	1/5	1/3	1/2	1/3	1/5	0.0498
Castle 2	5	1	3	4	1/3	2	0.2331
Castle 3	3	1/3	1	4	1/4	1/2	0.1156
Castle 4	2	1/4	1/4	1	1/5	1/3	0.0578
Castle 5	3	3	4	5	1	3	0.3805
Castle N	5	1/2	2	3	1/3	1	0.1633

Consistency ratio= 7.88 %

Table 24: Reciprocal matrix of alternatives by Security criterion and normalized priority vector

Security	Castle 1	Castle 2	Castle 3	Castle 4	Castle 5	Castle N	Normal priority vector
Castle 1	1	7	5	3	1	3	0.3550
Castle 2	1/7	1	2	1/2	1/2	1/3	0.0747
Castle 3	1/5	1/2	1	2	1/3	1/5	0.0715
Castle 4	1/3	2	1/2	1	1/3	1/2	0.0864
Castle 5	1	2	3	3	1	1	0.2143
Castle N	1/3	3	5	2	1	1	0.1981

Consistency ratio= 7.80 %

Table 25: Reciprocal matrix of the alternatives by access criterion and normalized priority vector

Access	Castle 1	Castle 2	Castle 3	Castle 4	Castle 5	Castle N	Normal priority vector
Castle 1	1	4	1/3	1/3	4	2	0.1740
Castle 2	1/4	1	1/3	1/3	1/3	1/5	0.0501
Castle 3	3	3	1	1	3	1	0.2399
Castle 4	3	3	1	1	5	3	0.3087
Castle 5	1/4	3	1/3	1/5	1	1/3	0.0692
Castle N	1/2	5	1	1/3	3	1	0.1582

Consistency ratio= 9.79 %

Table 26: Results of the global vector and ranking of the alternatives

Castle	Strength	Security	Access	Vector priority criteria	Normalized global vector	Ranking
Castle 1	0.04980	0.3550	0.1740	0.10852	0.2595	3
Castle 2	0.23309	0.0747	0.0501	0.54693	0.0834	6
Castle 3	0.11560	0.0715	0.2399	0.34454	0.1343	4
Castle 4	0.05777	0.0864	0.3087		0.1599	2
Castle 5	0.38047	0.2143	0.0692		0.1823	1
Castle N	0.16328	0.1981	0.1582		0.1806	5
Consistency ratio=	7.88 %	7.80 %	9.79 %	5.16 %		

Table 27: Results of the Unit Value of Castle N based on the Valuation Ratio

Castle	Unit Value [UM/m²]	Vvg [adim]	Valuation Ratio [UM/m²]	Vvg N [adim]	Unit Value N [UM/m²]
Castle 1	1,650.00	0.2595	11,672.56	0.1806	2.107.56
Castle 2	1,200.00	0.0834			
Castle 3	1,575.00	0.1343			
Castle 4	2,040.00	0.1599			
Castle 5	3,100.00	0.1823			
Total	9,565.00	0.8194		1.0000	

Table 28: Total value of Castle N

Castle	Construction area [m²]	Unit Value* [UM/m²]	Total Value [UM]
Castle N	1,500.00	2,100.00	3,150,000.00

*rounded

Table 29: Comparative table of the total value of the Castle N

Case	Unit Value [UM/m ²]	Total Value [UM]	Percentage difference
1	1,150.00	1,725,000.00	
2	2,100.00	3,150,000.00	82.61 %

«As can be seen, Your Highness, there is a significant difference of 82.61% between the two cases, which is sufficient evidence to conclude that the results of the global vector of alternatives are not binding on market prices. Although the AHP is correctly applied, the mathematical artifice does not guarantee reliable real estate market values. That's all, exclaimed the Wise Analytical. I now give way to the Wise Regressor to expound on his knowledge of the controversial procedure».



Magnanimous
King

Wise
Regressor

From the opinion of the Wise Regressor

For a greater empirical foundation on the inadvisability of using the AHP+RV in the real estate valuation, I expose to you, with all respect Your Highness, an analogous situation considering econometric models, which must follow these steps:

1. Define an economic function to explain real estate market prices.
2. To structure an econometric model considering the explained and explanatory variables.
3. Characterize the variables involved in the model.
4. To raise a market sample based on the variables that the researcher assumes are important in explaining real estate prices.
5. Econometric modeling using multiple linear regression models (MLRM).
6. Analyze the results and conclude on the point and interval estimation of the real estate value.

In order to fulfill the first step, the economic function is proposed:

$$PU=(AT,AC,CEN,CED,EC,FOR,SEG,ACC) \quad (4)$$

The regression equation is now posed:

$$Y_i = \beta_0 + \beta_1 AT_i + \beta_2 AC_i + \beta_3 CEN_i + \beta_4 CED_i + \beta_5 EC_i + \beta_6 FOR_i + \beta_7 SEG_i + \beta_8 ACC_i + \varepsilon_i, \text{ para } i=1...n \quad (5)$$

Table 30 - Explained and Explanatory Variables of the MRLM

Variable	Nom	Description	Unit	Type of Variable	Scale of Measurement
Land Area	AT	Total land area of the castle	ha	Explanatory Quantitative Continuous	Ratio
Construction Area	AC	Total construction area of the castle	m ²	Explanatory Quantitative Continuous	Ratio
Quality of Environment	CEN	Situation of the surroundings: Good: 1 Bad: 0 Quality Scale	Adm.	Explanatory Dummy Discrete	Ordinal
Quality of the building	CAD	Very good 4 Good 3 Regular 2 Bad 1 Conservation scale	Adm.	Explanatory Qualitative Discrete	Ordinal
State of Conservation	ECO	Very good 4 Good 3 Regular 2 Bad 1 Strength scale	Adm.	Explanatory Qualitative Discrete	Ordinal
Strength	CEN	Superior: 1 Normal: 0	Adm.	Explanatory Dummy Discrete	Ordinal
Security	CEN	Security scale Extreme: 1 High: 0 Access scale	Adm.	Explanatory Dummy Discrete	Ordinal
Access	ECO	Easy 3 Normal 2 Difficult 1	Adm.	Explanatory Qualitative Discrete	Ordinal
Unit Price	PU	Unit price of the offered castles	UM/m ²	Explained Quantitative Continuous	Ratio

Table 31- Values of the MRLM variables

Castillo	AT	AC	CEN	CED	ECO	FOR	SEG	ACC	PU	Castillo	AT	AC	CEN	CED	ECO	FOR	SEG	ACC	PU
1	20	1.500	1	2	1	0	1	1	1.650	36	25	1.531	1	2	4	0	1	1	2.332
2	15	2.500	1	1	2	1	0	2	1.575	37	36	1.615	0	1	2	0	1	2	2.535
3	30	3.500	0	3	3	0	0	3	2.040	38	21	2.626	1	4	2	1	1	2	1.815
4	20	4.000	0	4	4	1	1	2	3.100	39	36	3.100	0	1	2	1	0	1	2.451
5	20	2.792	0	2	1	0	1	3	2.298	40	21	3.842	0	3	2	0	1	2	2.130
6	17	3.202	1	3	2	0	1	3	2.469	41	22	3.172	1	4	1	1	1	3	2.320
7	34	2.256	1	3	4	0	0	3	2.878	42	37	2.307	0	3	2	1	1	1	2.291
8	23	1.678	1	3	1	0	1	3	2.089	43	18	2.602	1	3	4	0	1	2	2.025
9	29	1.993	1	4	3	0	0	3	1.849	44	29	2.381	1	2	1	0	1	3	2.676
10	39	1.814	0	2	2	1	1	2	2.679	45	40	2.876	1	4	3	0	0	1	3.047
11	17	1.855	1	1	2	0	1	2	2.560	46	36	3.667	0	4	3	1	0	1	1.663
12	19	2.487	0	4	3	1	0	1	2.946	47	24	3.195	0	1	1	1	1	3	2.158
13	17	2.062	0	4	1	0	0	2	2.037	48	34	3.976	0	3	1	1	0	2	2.537
14	28	2.693	1	3	3	1	1	2	2.213	49	31	3.459	0	3	2	1	1	2	1.788
15	36	2.046	1	1	1	1	1	1	2.348	50	26	3.478	1	2	3	0	1	1	2.479
16	20	3.204	1	2	2	0	0	3	2.225	51	31	2.566	1	4	2	0	0	2	2.661
17	18	2.963	0	1	1	0	0	3	2.799	52	17	3.993	0	3	3	0	0	2	2.280
18	25	2.643	0	4	2	0	1	3	2.304	53	35	2.334	1	2	4	1	1	2	2.758
19	20	3.186	0	4	3	0	0	3	1.846	54	29	2.867	0	3	2	1	0	1	2.400
20	16	3.177	1	4	4	1	0	2	2.713	55	22	2.625	1	2	4	1	0	1	2.295
21	18	3.102	0	4	1	0	1	1	2.436	56	38	2.093	0	4	3	1	0	3	1.688
22	23	1.718	1	2	4	1	0	2	1.727	57	20	1.774	1	3	2	0	1	2	2.303
23	26	2.082	1	3	3	1	0	2	2.856	58	31	3.789	0	1	2	1	0	2	2.293
24	22	2.967	0	3	3	0	1	3	2.488	59	28	2.463	0	2	1	1	1	1	1.750
25	39	2.043	1	4	2	0	1	2	2.429	60	35	3.363	1	3	1	1	1	3	1.653
26	39	1.933	0	2	1	0	1	1	2.158	61	24	1.916	1	4	4	1	0	3	2.711
27	22	3.668	1	2	1	1	0	2	2.680	62	29	1.585	0	2	4	0	1	2	2.424
28	25	2.603	0	2	3	0	0	2	2.247	63	32	3.440	0	2	2	1	1	2	2.490
29	31	3.130	1	1	2	1	0	1	1.949	64	30	3.767	1	3	2	1	1	2	2.486
30	38	3.763	1	1	1	1	0	3	2.749	65	24	2.627	1	4	4	1	1	1	1.755
31	29	2.277	0	4	1	1	0	2	2.382	66	29	3.353	1	2	2	1	1	2	2.647
32	22	2.727	0	4	1	1	0	2	1.827	67	32	1.715	1	3	1	0	0	3	2.583
33	35	3.852	1	4	4	0	0	3	2.720	68	30	3.540	1	1	2	1	1	3	2.781
34	37	1.644	1	2	2	1	0	1	2.894	69	34	3.485	0	4	2	0	1	1	2.182
35	15	2.933	0	1	2	1	0	2	2.658										

From the random database

$$PU_{est} = +1.899,28 + 8,71 * AT + 0,07 * AC + 82,63 * CEN - 50,73 * CED + 51,37 * ECO - 76,59 * FOR - 59,95 * SEG + 29,02 * ACC$$

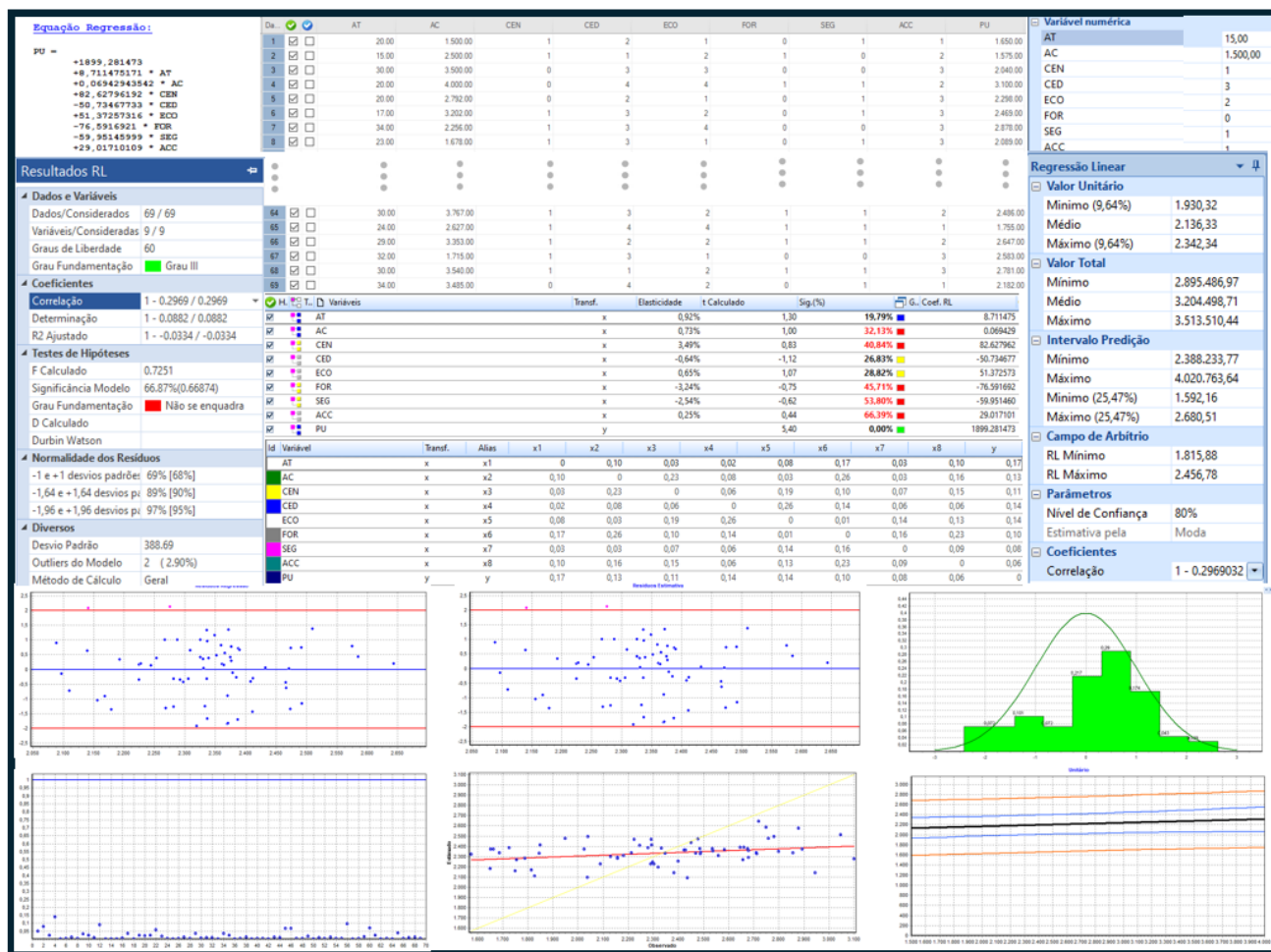


Table 32: Results of the MRLM test statistics

Statistic	Value	Observations
Correlation coefficient (r)	0,30	A weak correlation between the explained variable and the explanatory variables.
Coefficient of determination (R^2)	0,0882	8.82 % of the variation in unit values around the mean is explained by the independent variables included in the model. 91.18 % of the variations are unexplained.
Calculated F	0,7251	The level of significance of the model is 66.87 %, which is greater than the significance level α , so the null hypothesis is accepted, therefore the independent variables included in the model are not important for explaining the variability of prices observed in the market.
Significance of population estimators	19,79 - 66,39 %	The regressors of the variables are not statistically significant for the Level III Foundation.
Standard deviation	388,69	Represents 18.19% of the estimated mean value of the property.
Outliers	2,90 %	Within the tolerance of < 5%.
Cook's distance	0	No presence of influential points.

Given these results, the MRLM is rendered invalid without the need to assess further basic assumptions or model specification errors.

Table 33: Total value of taxable property (Castle N)

Castle	Construction area [m ²]	Unit Value* [UM/m ²]	Total Value [UM]
Castle N	1,500.00	2,200.00	3,300,000.00

*Rounded

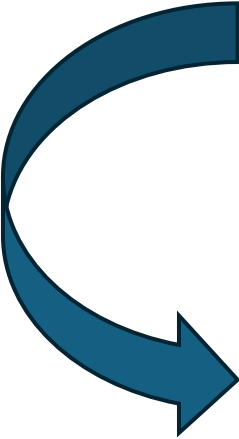
«This result obtained does not have a valid empirical basis as was explained in the presentation of the MRLM calculations, as well as the one determined using the AHP+RV (3,150,000.00 UM). In spite of a percentage difference of +4.76% of the Value using the MRLM with respect to the Value using AHP + RV, it is a mere coincidence of results, both are biased, thus proving that the results of the global vector of the criteria are not binding with the prices of the real estate market, even if the AHP method is well applied, the Valuation Ratio as a mathematical artifice does not guarantee market real estate values».



From the verification of the method
of *reductio ad absurdum*

Alternative hypothesis (contrary):

**The results of the general weighting vector of the AHP
are linked to real estate market prices**



Contradiction or absurd result



Initial hypothesis (original):

**The results of the general weighting vector of the AHP
are not linked to real estate market prices.**



Discussion and conclusion



Discussion:

- There is no doubt about the effectiveness of AHP in decision making. But the valuers have the obligation to consult the original sources of knowledge, never accept a methodological procedure as valid, there is always a need to prove its empirical foundation because that is where the discovery of information enhances the investigation into the science of valuation.

Conclusion:

- Based on the ad absurdum argumentative strategy, and with the results of the investigation it was demonstrated that **the alternative hypothesis is false**, therefore, the results of the general weighting vector of the AHP **are not linked** to the prices of the real estate market. Therefore, the valuation ratio as a mathematical artifice does not guarantee market real estate values.



Conference

**Why are AHP results not binding
for appraising real estate?**

Screenplay and editing

Miguel Alberto Camacaro

Música

Pixabay.com

Image

Léxica.art/Microsoft designer

Voice

Eleven Labs

Audio

www.audacityteam.org

Avatars and videos

<https://aiapp-es.vidnoz.com/>

Base program recording

microsoft-365/p/powerpoint

Recording program

obsproject.com/es

Edition program

www.vegascreativesoftware.com/es

Studio

Torrejón Films